Portfolio Theory

Professor Philip Howard
howardpd@wfu.edu

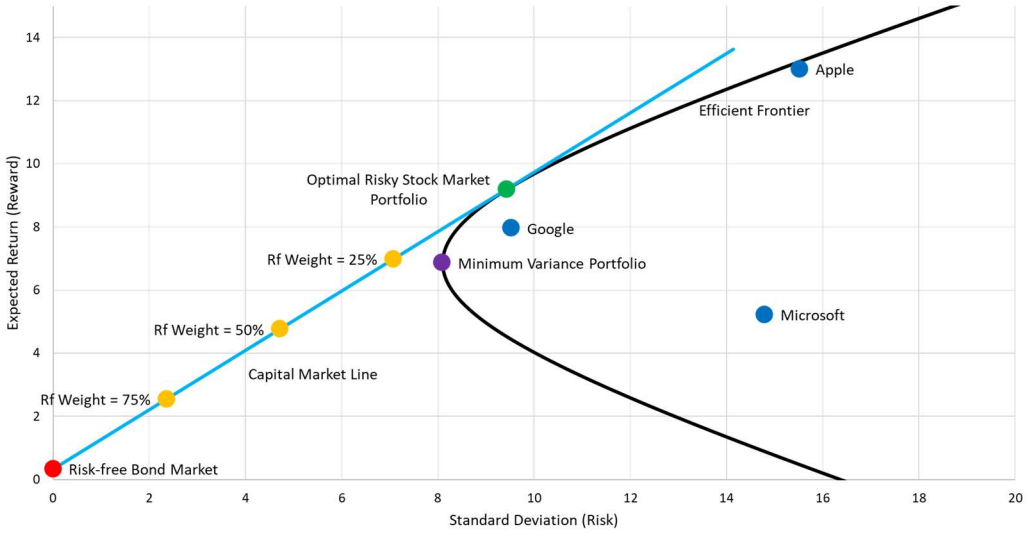
WAKE FOREST UNIVERSITY
SCHOOL of BUSINESS

Portfolio Theory

- Given a menu of assets, what do we invest in and how much?
- How do we define best? Sharpe ratio
- Choose investment weights w_i to maximize portfolio Sharpe (Reward-to-Risk) ratio:

$$\max \frac{E[R^P] - R^f}{\sigma_P}$$

Portfolio Optimization



Leveraged Funds

- Funds use leverage to enhance returns

ProShares UltraPro QQQ (TQQQ)

- "Seeks daily investment results, before fees and expenses, that correspond to three times (3x) the daily performance of the Nasdaq-100 Index."
- https://www.proshares.com/funds/tqqq.html

Leveraged Funds - ProShares UltraPro QQQ

130-30 Funds

- Funds allow limited short positions (130% long / 30% short) to enhance returns

JPMorgan U.S. Large Cap Core Plus Fund (JLPSX)

- "Attempts to generate additional returns over the S&P 500 Index and long-only strategies through the ability to sell stocks short"
- https://am.jpmorgan.com/us/en/asset-management/gim/adv/products/d/jpmorgan-u-s-large-cap-core-plus-fund-i-4812a2389#/overview

Portfolio Stats

Number of Holdings As of 08/31/2020	292
Fund Assets As of 08/31/2020	$3.41bn
Gross Exposure (Long + Short)	152.17%
Net Exposure (Long - Short)	100.11%
Turnover Ratio	
Turnover Ratio - Long (Trailing 12 month) (6/30/2020)	85.00%
Turnover Ratio - Short (Trailing 12 month) (6/30/2020)	134.00%
Wtd. Avg. Market Cap	$545.82bn
Median Market Cap	$28.73bn
P/B Ratio	4.42
P/E Ratio (1 yr. forecast)	25.67
P/E Ratio (1 yr. trailing)	26.89
Percentage of holdings	
LONG	126.14%
SHORT	26.03%

Minimum Volatility Funds

- Funds seek to minimize volatility while tracking benchmark

Vanguard Global Minimum Volatility Fund Admiral Shares (VMNVX)

- "The fund invests primarily in U.S. and foreign stocks that are expected to minimize volatility relative to the global equity market. To reduce volatility, the fund will consider stocks' risk and diversification characteristics."
- https://investor.vanguard.com/mutual-funds/profile/overview/vmnvx

Minimum Volatility Funds - Vanguard Global Minimum Volatility Fund

Risk Management

- Banks and insurance companies will allocate across a portfolio of holdings to minimize downside risk

Value at Risk (VaR)

- If asset has 5% VaR of $1 million, there is a 5% probability of the asset losing $1 million or more

Forecasts

- Historical data is one source of information available to you
- Forecasts is a second source
  - Analyst forecasts
  - Economic models
  - Your own business expertise

- You can use your forecasts to also perform portfolio optimization

Random Variables

- Random variable has random outcomes
- Outcomes have probabilities
- Two types of probability distributions
  - Discrete: outcomes are discrete values
  - Continuous: outcomes over continuous interval
  - Probability distributions sum to 1

Expected Value

Let X be a discrete random variable with n outcomes

- X_i the i-th outcome
- $P(X_i)$ the probability of X_i

- Expected value

$$E[X] = \sum_{i=1}^{n} P(\mathrm{X}_i) \times \mathrm{X_i}$$

Variance & Standard Deviation

- Variance

$$Var(R) = \sigma^2 = \sum_{i=1}^{n} P(X_i) \times (X_i - E[X])^2$$

- Standard Deviation

$$SD(R) = \sigma = \sqrt{Var(R)} = \sqrt{\sigma^2}$$

Covariance & Correlation

- Covariance

$$Cov(X_1, X_2) = \sigma_{1,2} = \sum_{i=1}^{n} Prob_i \times \left(X_{1,i} - E[X_1]\right) \times \left(X_{2,i} - E[X_2]\right)$$

- Correlation

$$Corr(R_1, R_2) = \rho_{1,2} = \frac{\sigma_{1,2}}{\sigma_1 \sigma_2}$$